CS 3244

Machine Learning

Week 5 – Bias and Variance & Overfitting

5 post

Kan Min-Yen

Background Photo credits: Rafiq Mirza, Luan Ahn, Rosmarie Voegtli @ Flickr

Recap and Forecast

Linear Regression

- Real valued output

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

Logistic Regression

- Probability of a binary event

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + e^{-y^{(j)}\theta^\top x^{(j)}})$$

Gradient Descent

- Take steps in opposite the gradient of the cost function.
- SGD: the average pointwise direction is equivalent to the gradient $-\nabla J$

Learning Outcomes for this week

- Understand the bias–variance tradeoff
- Understand why overfitting occurs and the role of the model complexity and sampled dataset in causing it
- Understand the effect of noisy targets as contributing another source of noise
- Be clear about the role of data resources towards model selection

Bias and Variance

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Approximation–generalization tradeoff

Small J_{test}: good approximation of f on unseen test data

More complex $\mathcal{H}$ → better chance of **approximating** f

Less complex $\mathcal{H}$ → better chance of **generalizing** out of sample

Ideal $\mathcal{H} = \{f\}$

Good luck with that!

Quantifying the tradeoff

Decomposing J_{test} into

1. How well $\mathcal{H}$ can approximate f overall
2. How well we can zoom in on a good $h_\theta \in \mathcal{H}$

Applies to **real-valued** targets and uses **squared error**

There is an equivalent binary version through the lens of VC dimensions

Start with J_{test}

$$J_{test}(h_{\mathcal{D}}) = \mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]$$

$h(x)$ depends on the specific dataset $\mathcal{D}$.

Start with J_{test}

$$J_{test}(h_{\mathcal{D}}) = \mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]$$

$h(x)$ depends on the specific dataset $\mathcal{D}$.

$$\mathbb{E}_{\mathcal{D}}[J_{test}(h_{\mathcal{D}})]$$

Generalizing over all $\mathcal{D}$ with same m

$$= \mathbb{E}_{\mathcal{D}}\left[\mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

$$= \mathbb{E}_x\left[\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

Swap ok, as integrand is strictly non-negative

Focus on $\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]$, get the outer term $\mathbb{E}_x[\ldots]$ later.

The average hypothesis

To evaluate $\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]$,

we define the 'average' hypothesis $\bar{h}(x) = \mathbb{E}_{\mathcal{D}}[h_{\mathcal{D}}(x)]$

Imagine many, many data sets $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K$ drawn:

$$\bar{h}(x) \approx \frac{1}{K} \sum_{k=1}^{K} h_{\mathcal{D}}(x)$$

Using $\bar{h}(x)$

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] =$$

Using $\bar{h}(x)$

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]$$

$$= \mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x) + \bar{h}(x) - f(x)\right)^2\right]$$

Add $-\bar{h}(x) + \bar{h}(x)$. Associate first two and second two terms.

$$= \mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2 + \left(\bar{h}(x) - f(x)\right)^2 + 2\underbrace{\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)}_{0}\underbrace{\left(\bar{h}(x) - f(x)\right)}_{\text{const w.r.t. } \mathcal{D}}\right]$$

Expand out, cross terms drop as first part of term is 0.

$$\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right] + \left(\bar{h}(x) - f(x)\right)^2$$

2nd term is constant with respect to $\mathcal{D}$

Bias and Variance

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{①} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{②}$$

Quick Quiz: Which is which?

Bias and Variance

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}}$$

Therefore,

$$\mathbb{E}_{\mathcal{D}}[J_{test}(h_{\mathcal{D}})] = \mathbb{E}_x\left[\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

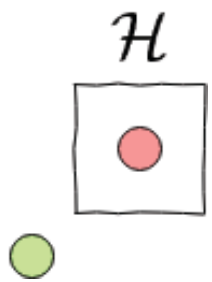
$$= \mathbb{E}_x[\text{bias}(x) + \text{var}(x)]$$

$$= \text{bias} + \text{var}$$

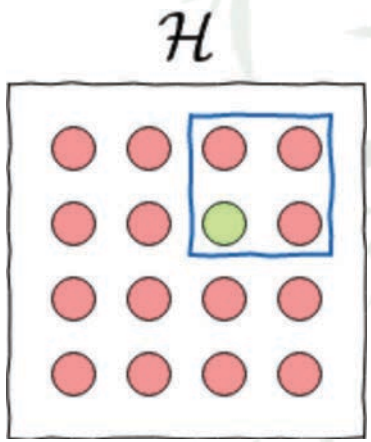
The tradeoff

$$\text{Bias}[h(x)] = \left|\bar{h}(x) - f(x)\right|$$

$$\text{Var}[h(x)] = \mathbb{E}\left[\left(h(x) - \bar{h}(x)\right)^2\right]$$



$\mathcal{H}_{small}$: just one hypothesis.



$\mathcal{H}_{big}$: big, contains the target function f. But you have to find it.

Overfit Measure

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

An irony of two learners

Two learners Overfitting and Restrained:

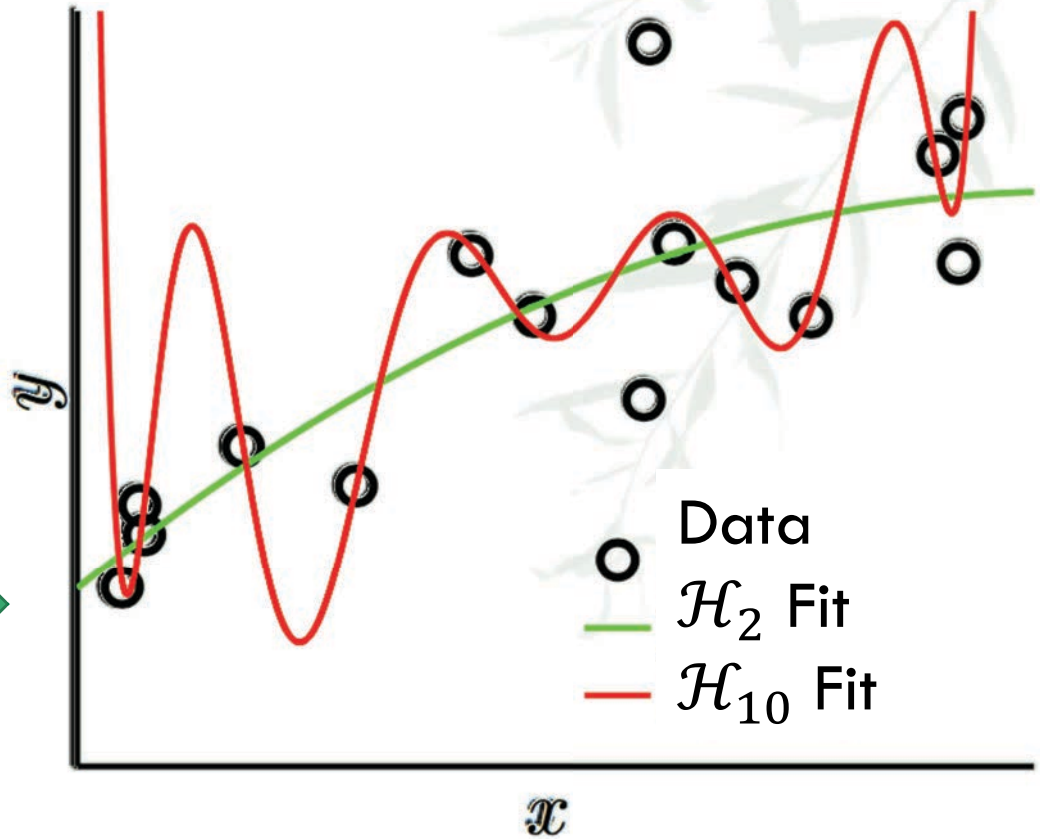
Both <u>know</u> the target is 10th order,

you get 15 data points.

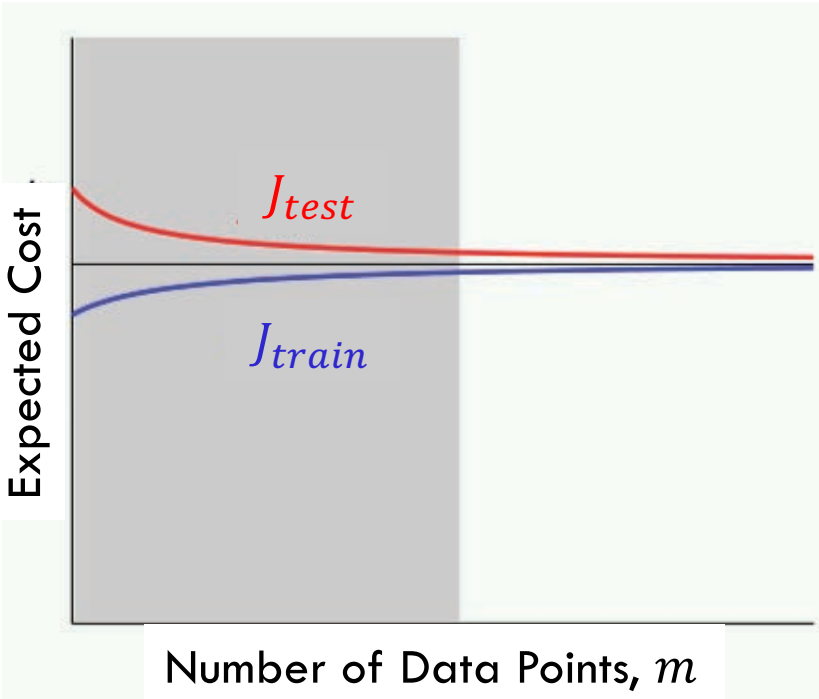
O chooses $\mathcal{H}_{10}$

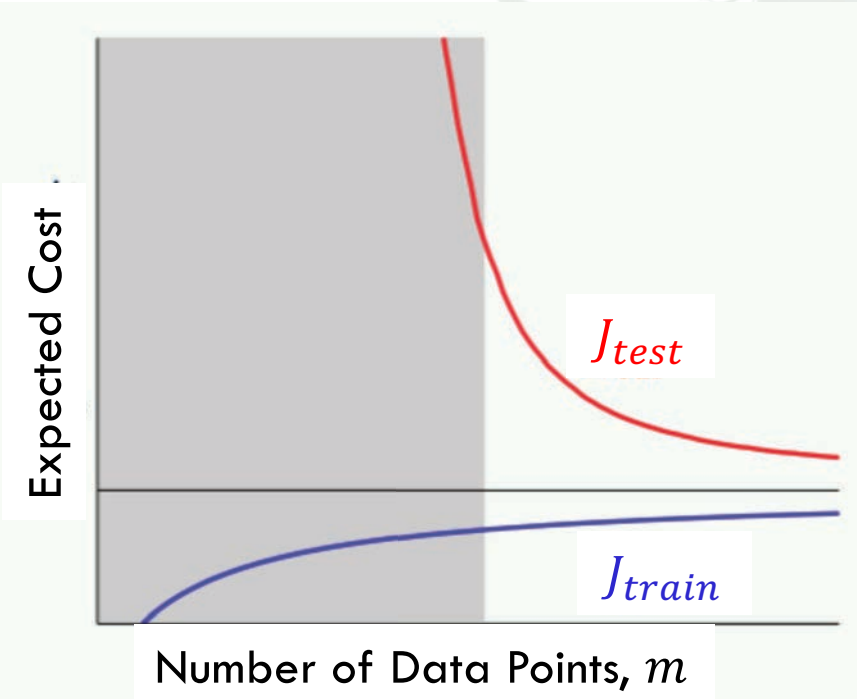
R chooses $\mathcal{H}_2$

Learning a 10th order target with noise



When is $\mathcal{H}_2$ better than $\mathcal{H}_{10}$?





Overfitting (grey region): $J_{test}(\mathcal{H}_{10}) > J_{test}(\mathcal{H}_2)$

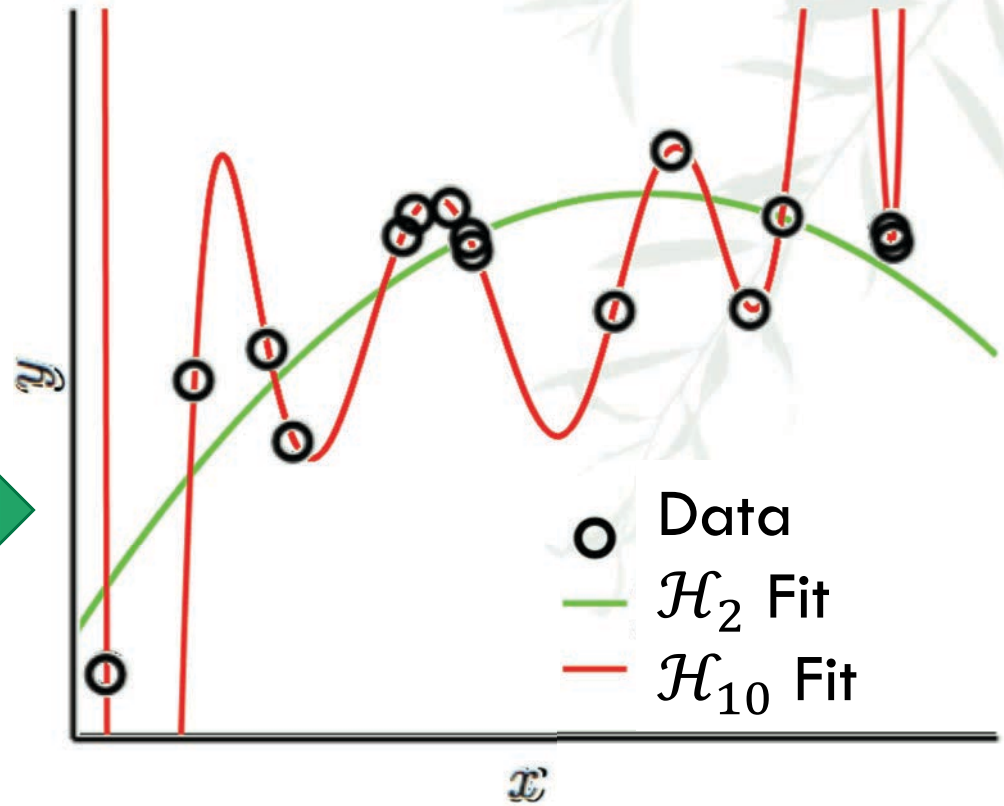
Even without noise with complex f

The two learners $\mathcal{H}_{10}$ and $\mathcal{H}_2$

They <u>know</u> there is no noise

Is there really no noise?

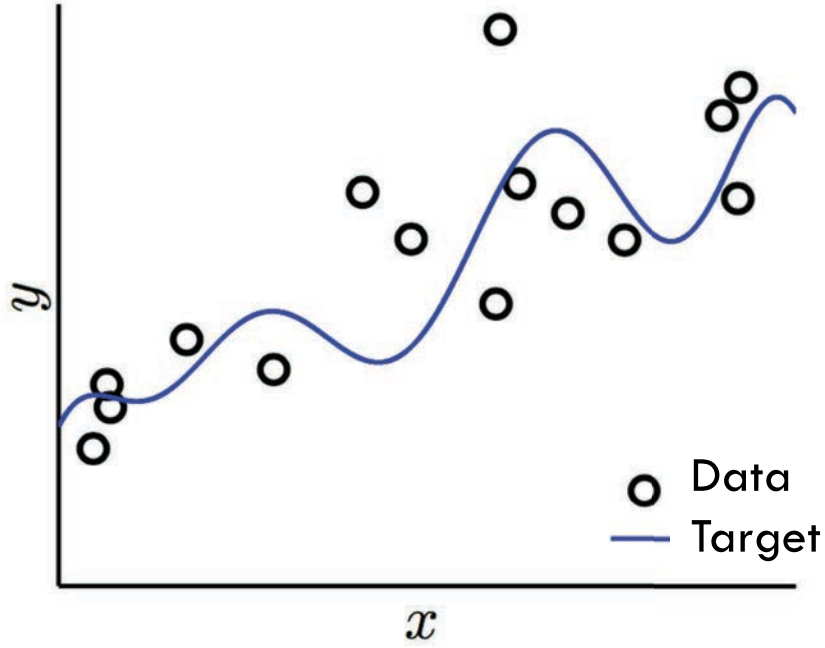
Learning a 50th order target



A detailed experiment

Impact of **noise level** and **target complexity**

$$y = f(x) + \epsilon(x) = \underbrace{\sum_{q=0}^{Q_f} \alpha_q x^q}_{*} + \underbrace{\epsilon(x)}_{\sigma^2}$$



Noise level: σ^2

Target complexity: Q_f

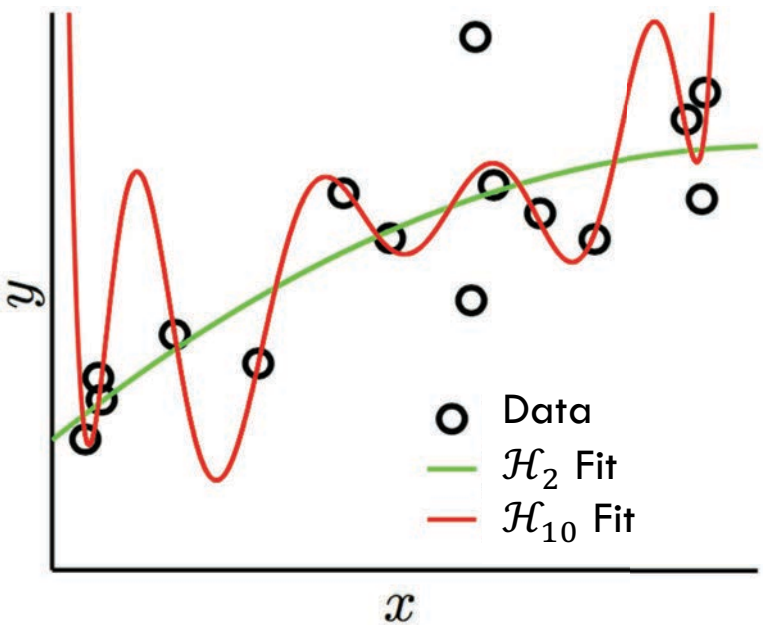
Data set size: m

The overfit measure

We fit the data set $(x_1, y_1), \ldots, (x_m, y_m)$ using our 2 models:

$\mathcal{H}_2$: 2nd order polynomials

$\mathcal{H}_{10}$:10th order polynomials



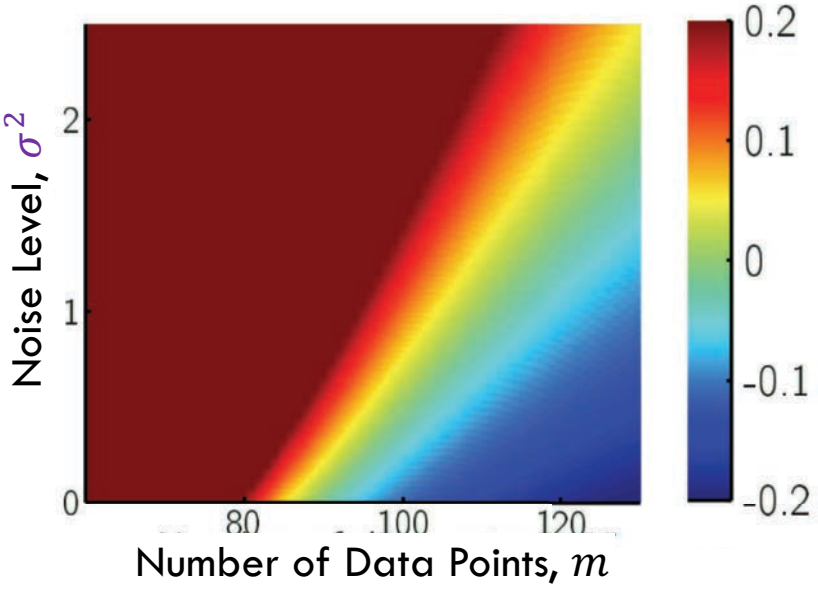
Compare out-of-sample errors of $h_2 \in \mathcal{H}_2$ and $h_{10} \in \mathcal{H}_{10}$

Overfit measure:

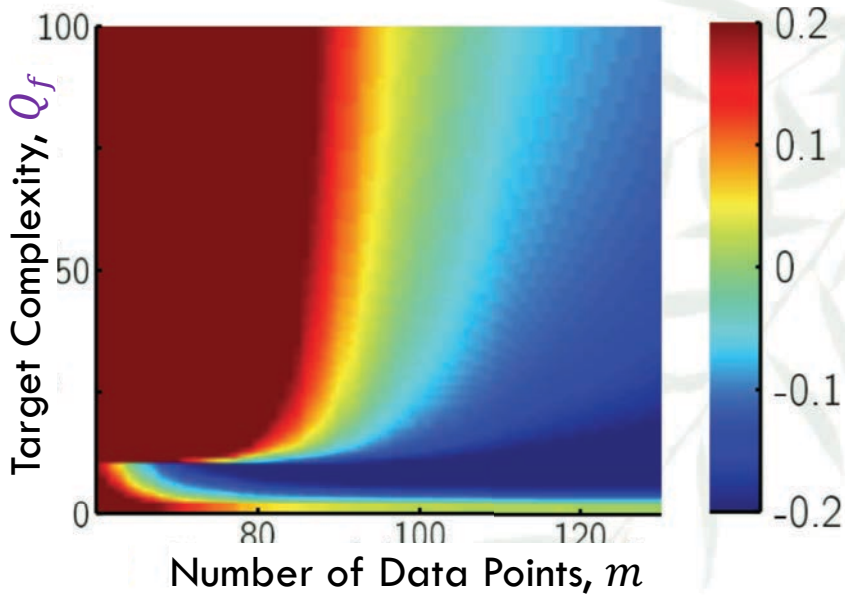
$$J_{test}(h_{10}) - J_{test}(h_2)$$

Overfit measure:

$$J_{test}(h_{10}) - J_{test}(h_2)$$



Stochastic Noise



Deterministic Noise

Number of data points	↑	Overfitting ↓
Stochastic Noise	↑	Overfitting ↑
Deterministic Noise	↑	Overfitting ↑

The role of Noise

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Noise

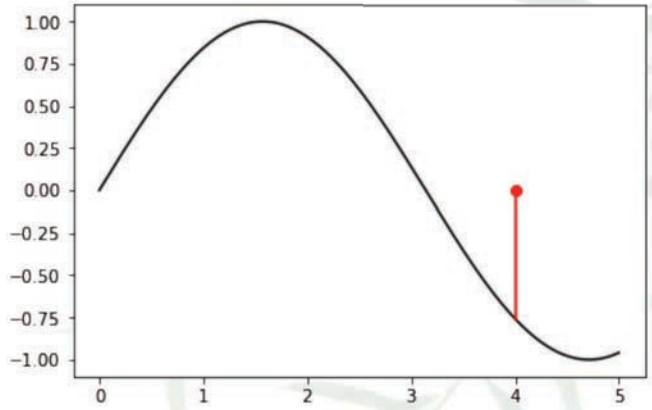
That part of y that we **cannot** model

It has two sources …

Stochastic Noise: Data Error

We would like to learn f

$$y_j = f(x_j)$$

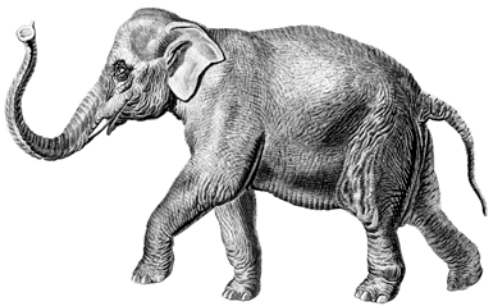


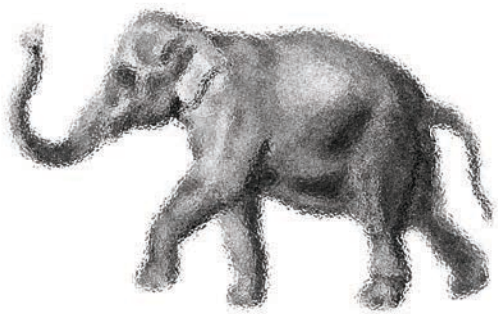
Unfortunately, we only observe $\mathcal{D}$

$$y_j = f(x_j) + \texttt{stochastic_noise}$$

Stochastic Noise

Fluctuations that we cannot model



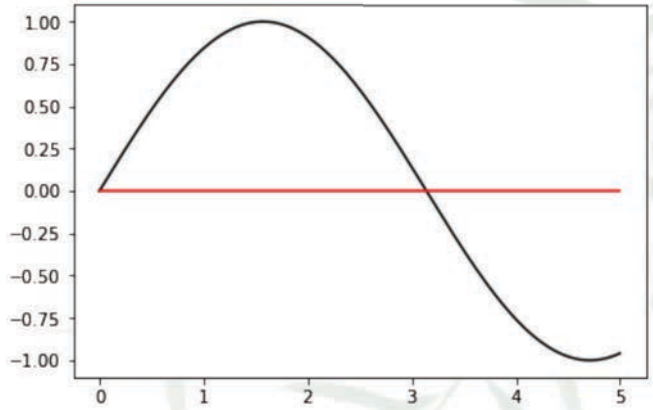


images of elephant from rawpixel.com

Deterministic Noise: Model Error

We would like to learn from O

$$y_j = \bar{h}(x_j)$$

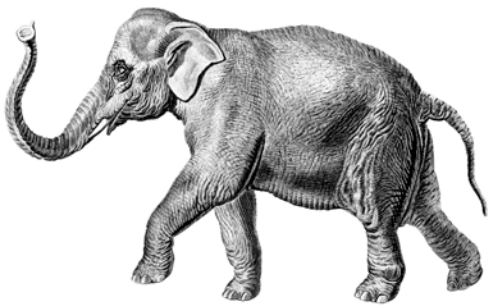


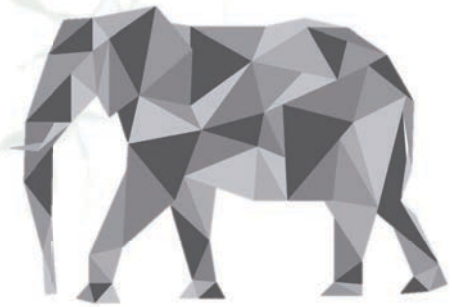
Unfortunately, we only observe $\mathcal{D}$

$$y_j = f(x_j)$$
$$= \bar{h}(x_j) + \texttt{deterministic_noise}$$

Deterministic Noise

The part of f we lack the capacity to model

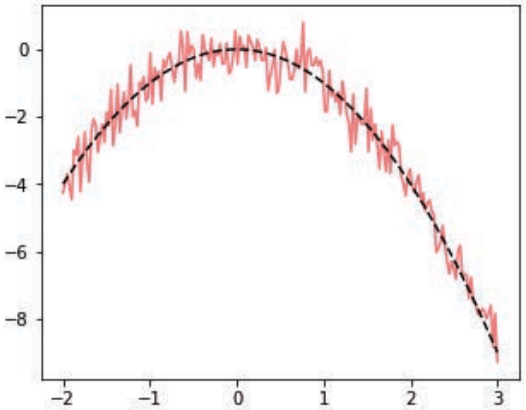




images of elephant from rawpixel.com

Both sources of noise hurt learning

Stochastic Noise

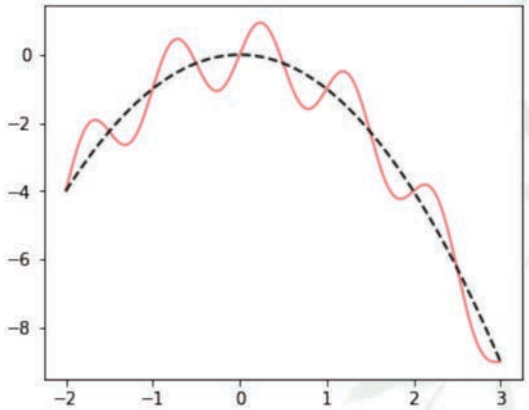


Source: random measurement errors

Re-measure y: Stochastic noise changes

Change $\mathcal{H}$: Stochastic noise still the same

Deterministic Noise



Source: learner's $\mathcal{H}$ cannot model f

Re-measure y: Deterministic noise the same

Change $\mathcal{H}$: Deterministic noise changes

We have a single $\mathcal{D}$ and fixed $\mathcal{H}$ so we cannot distinguish between either; with finite m, $\mathcal{H}$ will try to fit the noise.

Noise and Bias-Variance

Recall the decomposition:

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}}$$

What if f is a noisy target?

$$y = f(x) + \epsilon(x) \qquad \mathbb{E}[\epsilon(x)] = 0$$

A noise term

$\mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - y)^2] =$

A noise term

$$\mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - y)^2]$$

$$= \mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - f(x) - \epsilon(x))^2]$$

Expand observed y

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(\underbrace{h_{\mathcal{D}}(x) - \bar{h}(x)}_{①} + \underbrace{\bar{h}(x) - f(x)}_{②} - \underbrace{\epsilon(x)}_{③}\right)^2\right]$$

Associate terms

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2 + \left(\bar{h}(x) - f(x)\right)^2 + (\epsilon(x))^2 + \cancel{\text{cross terms}}\right]$$

Additional cross terms disappear as $\mathbb{E}(\epsilon(\boldsymbol{x})) = 0$

Actually two noise terms

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] =$$

$$\underbrace{\mathbb{E}_{\mathcal{D},x}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\mathbb{E}_x\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}} + \underbrace{\mathbb{E}_{\epsilon,x}\left(\epsilon(x)\right)^2}_{\sigma^2}$$

Deterministic Noise

Stochastic Noise

K-nn Regression with Noise

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

k-nearest Regression with Noise

Assume $y^{(j)} = f(x^{(j)}) + \epsilon$ where $\mathbb{E}(\epsilon) = 0$ & $Var(\epsilon) = \sigma^2$.

Using squared error loss, the expected prediction error of a regression fit $f(x)$ at an input point $x = x^{(0)}$ can be written as the sum of Stochastic error, Bias^2 and Variance.

For the k-nearest regression fit, the error can be expressed as:

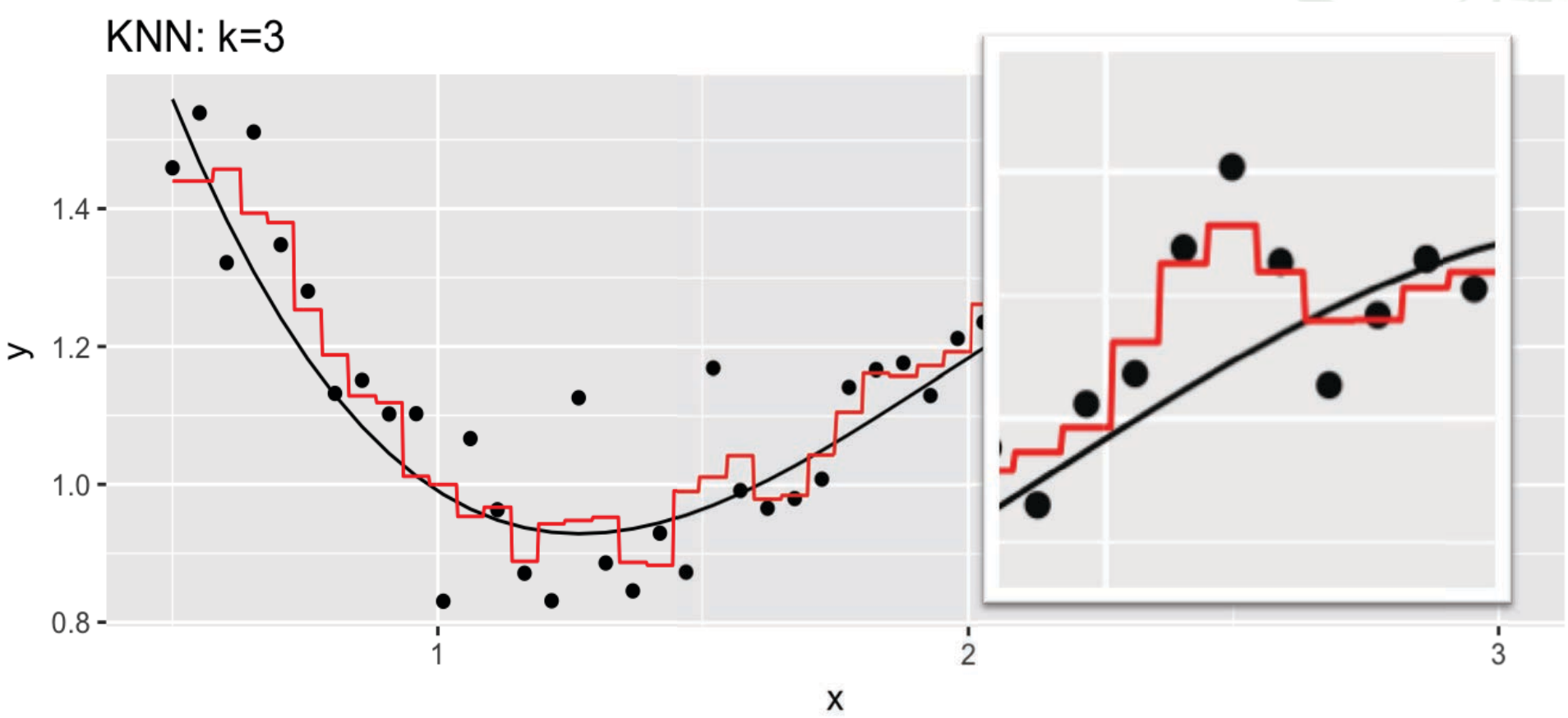
$$J(x^{(0)}) = \mathbb{E}[\left(y^{(0)} - h\left(x^{(0)}\right)\right)^2 |x^{(0)})]$$

$$= \sigma^2 + (f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f(x^{(j)}))^2 + \frac{\sigma^2}{k}$$

where $x^{(j)}$ $(j = 1, \dots, k)$ are the k nearest data points.

Derive the above components of the equation.

kNN Regression in action



Derivation

- Stochastic Error σ^2

- Bias^2 $\left(f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)^2\right.$

- Variance $\frac{\sigma^2}{k}$

Bias^2

Bias^2

$$\left(y\left(x^{(0)}\right) - \mathbb{E}_{\mathcal{D}}\left[h_k\left(x^{(0)}\right)\right]\right)^2 = \left(y\left(x^{(0)}\right) - \mathbb{E}_{\mathcal{D}}\left[\frac{1}{k}\sum_{j=1}^{k} y\left(x^{(j)}\right)\right]\right)^2$$

$$= \left(y\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} y\left(x^{(j)}\right)\right)^2$$

Assuming fixed neighbors $x^{(j)}$,

$$\mathbb{E}_{\mathcal{D}}\left[\frac{1}{k}\sum_{j=1}^{k} y\left(x^{(j)}\right)\right] = \frac{1}{k}\sum_{j=1}^{k} y\left(x^{(j)}\right)$$

$$= \left(f\left(x^{(0)}\right) + \epsilon - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right) + \epsilon\right)^2$$

Over all $\mathcal{D}$, the ϵ values disappear because they have zero mean

$$= \left(f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)\right)^2$$

Variance

Variance

$$Var(h_k(x^{(0)}) = Var(\frac{1}{k}\sum_{j=1}^{k} y(x^{(j)}))$$

$$= \frac{1}{k^2}\sum_{j=1}^{k} Var(f(x^{(j)}) + \epsilon^{(j)})$$

$Var(aX) = a^2 Var(X)$, as $\mathbb{E}(aX) = a\mathbb{E}(X)$

$$= \frac{1}{k^2}\sum_{j=1}^{k} Var(f(x^{(j)})) + Var(\epsilon^{(j)})$$

$$= \frac{1}{k^2}\sum_{j=1}^{k} Var(\epsilon^{(j)})$$

$Var(f(x^{(j)})) = 0$, because we assume the neighbors $x^{(j)}$s are fixed, and hence have no variance.

$$= \frac{1}{k^2}\sum_{j=1}^{k} \sigma^2 \quad = \frac{\sigma^2}{k}$$

Learning Curves

NUS School of Computing

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Learning Curves

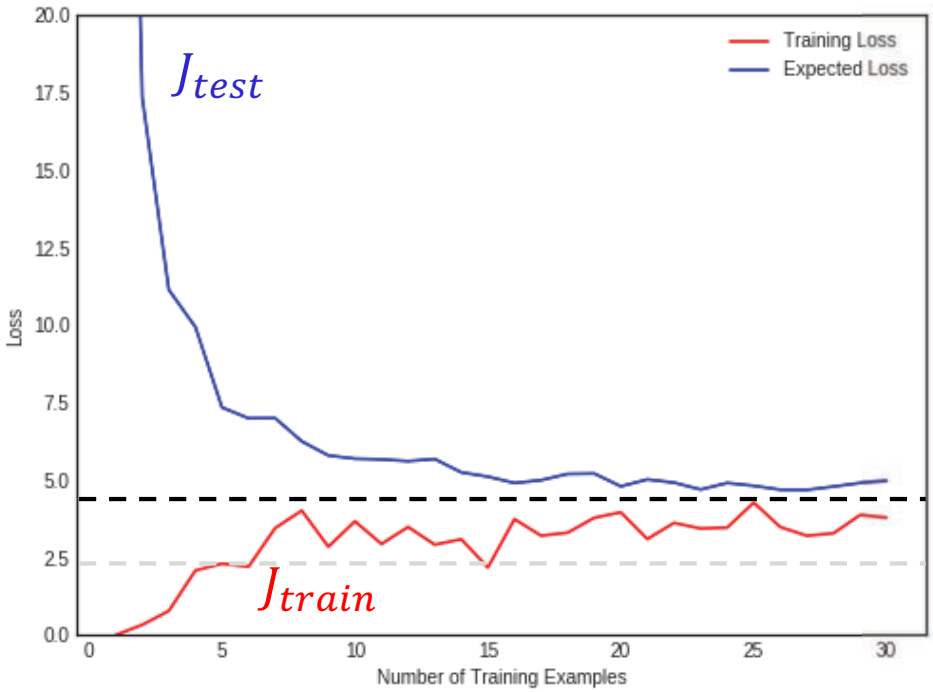
Plot expected J_{test} and J_{train} …

…as we vary size m

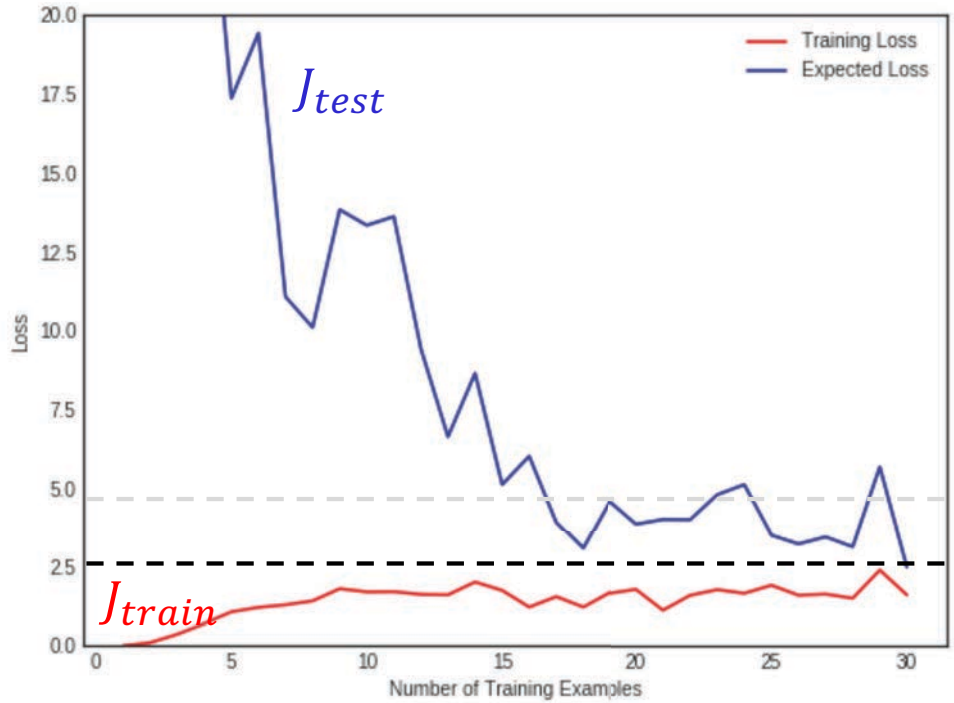
Expected test cost $\mathbb{E}_{\mathcal{D}}[J_{test}(f_{\mathcal{D}})]$

Expected training cost $\mathbb{E}_{\mathcal{D}}[J_{train}(f_{\mathcal{D}})]$

Simple versus Complex Models

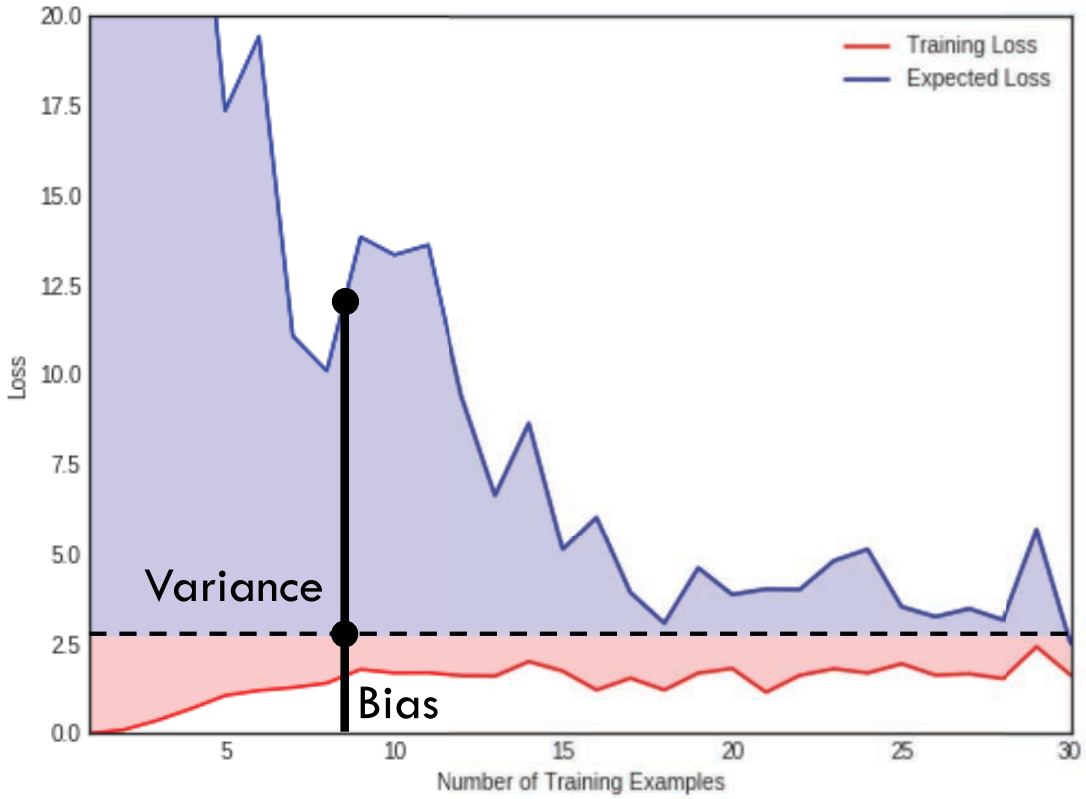


Simple Model



Complex Model

Bias–Variance on Curve



Prelude:

Curing Overfitting

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Noise causes overfitting

Overfitting is the disease

Noise is the cause

Learning is led astray by fitting the noise more than the signal

Two Cures:

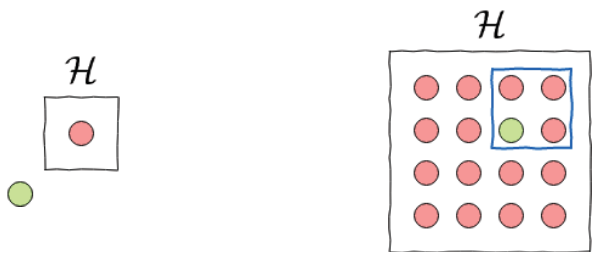
1. **Regularization**: Restrain the model
2. **Validation**: Reality check by peeking at (the bottom line)

Summary

Bias-Variance Tradeoff

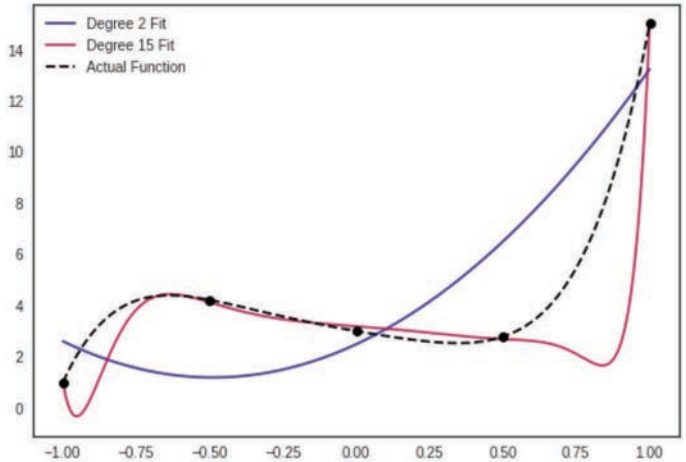
<u>Bias</u>: How well $\mathcal{H}$ can approximate f overall

<u>Variance</u>: How well we can zoom in on a good $h \in \mathcal{H}$



Match the 'model complexity' to the **data resources**, not to the target complexity

Overfitting: Fitting the data more than is warranted



Two causes: fitting stochastic + deterministic noise

Bias $\equiv$ deterministic noise